RR-S-1936
August 28, 1975
SUPERSEDING
MIL-S-10897D
October 30, 1969

FEDERAL SPECIFICATION

SCOOPS, ICE CREAM, MECHANICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of mechanical ice cream scoop (see 6.1).

1.2 Classification. The ice cream scoops shall be of the following sizes as specified (see 6.2).

Size 6 - 6 scoops per quart.
Size 8 - 8 scoops per quart.
Size 12 - 12 scoops per quart.
Size 20 - 20 scoops per quart.
Size 30 - 30 scoops per quart.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-P-533	-	Paper, Wrapping, Tissue.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-665	-	Boxes: Paperboard, Metal Edged and Components.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

FSC 7330

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Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of in invitation for bids or request for proposal shall apply:

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American Society for Testing and Materials (ASTM)

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

A276 - Stainless and Heat-Resisting Steel Bars, Hot-Finished or Cold-Finished

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Sanitation Foundation (NSF)

Criteria C-2 - Special Equipment and/or Devices

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

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3. REQUIREMENTS

3.1 Standard product. The ice cream scoops delivered under this specification shall be the manufacturers' commercially current standard product, except for any changes necessary to comply with specification requirements.

3.2 Standard compliance. The ice cream scoops shall conform to the applicable requirements of NSF Basic Criteria No. C-2.

3.2.1 NSF certification. The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the ice cream scoop he proposes to furnish under this specification meets the applicable requirements of Criteria C-2 and shall be:

(1) A listing in the current edition of NSF "Listing of Food Service Equipment" or display of the NSF Seal on the finished ice cream scoop, or

(2) A certification for the ice cream scoop issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the ice cream scoops are being procured, indicating that the ice cream scoops have been tested and conform to Criteria C-2.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his ice cream scoop, provided the completed item complies with all provisions of this specification. Commercial tolerances are applicable to all material thicknesses and diameters.

3.3.1 Steel, corrosion-resisting.

3.3.1.1 Bar. Corrosion-resisting round steel bar shall conform to the chemical composition of type 302, 303, 304, or 430 of ASTM Specification A276.

3.3.1.2 Sheet and strip. Corrosion-resisting steel sheet and strip shall conform to the chemical composition of type 302, 303, or 304 of ASTM Specification A167.

3.4 Design and construction. The ice cream scoops shall be designed as specified herein and as illustrated in figure 1. Figure 1 is illustrative only except for dimensions. Each size scoop shall consist of a bowl, bowl knife assembly, handle assembly, and spring-loaded thumb pad for operating the knife assembly.

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3.4.1 Bowl. The bowl shall be fabricated from 0.025-inch minimum-thick corrosion-resisting steel specified in 3.3.1.2. The bowl shall be attached to the shank (see 3.4.3) by spot welding, seam welding, or silver soldering. Each size bowl shall have the capacity listed in table I when tested as specified in 4.3.1. When spot welded, the interface between the bowl and shank shall be sealed 360 degrees with solder, silver solder or plastic resin. Welded joints shall not fail when tested as specified in 4.3.3.

TABLE I

Size	Capacity (fluid ozs.) ± 5 percent	Color code (handle end)
6	5	white
8	4	grey
12	2-2/3	orange
20	2	yellow
30	1	black

3.4.2 Bowl knife assembly. The bowl knife assembly shall consist of a shaft, knife, and gear, and shall be either one piece construction or fabricated as separate pieces. When fabricated of separate pieces, the knife in the bowl shall be mechanically locked to a shaft assembly and shall be removable without the use of tools. The shaft shall have a flat or square end and shall fit a matching hole in the gear. The gear shall be securely staked, welded, or soldered to the shaft to prevent longitudinal movement. When pressure is applied to the thumb pad, the gear, shaft, and knife shall rotate against the action of a torsion spring. The knife, thumb pad, gear, and quadrant shall be fabricated from corrosion-resisting steel specified in 3.3.1.2. The shaft shall be fabricated from corrosion-resisting steel specified in 3.3.1.1. The spring shall be fabricated from corrosion-resisting steel wire. The knife shall rotate 180 degrees through the empty bowl and be returned to its original position by the action of the torsion spring without binding or malfunctioning, when tested as specified in 4.3.2. The knife shall fit the inside contour of the bowl and shall have a clearance of 0.045 inch maximum at the equator and 0.030 inch maximum clearance at the center of the bowl bottom.

3.4.3 Handle assembly. The handle assembly shall consist of a handle, shank, guard, and color plug as illustrated on figure 1. The handle shall be constructed from black, thermosetting plastic, nylon, polypropylene, molded or threaded to a shank. The shank shall be fabricated from corrosion-resisting steel specified in 3.3.1.1. The guard shall be fabricated from

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corrosion-resisting steel specified in 3.3.1.2. The grip of the handle shall be grooved on at least half the circumference and shall have a colored plastic plug as specified in table I, located in the base of the handle to identify the size of the scoop. The handle assembly shall not permanently deform, crack or fail when tested as specified in 4.3.3.

3.5 Finish. All exposed metallic surfaces shall be smooth and shall be free from discoloration, rust, or stains. The molded handle shall be smooth.

3.6 Marking. Each ice cream scoop shall be permanently marked with the manufacturer's name, trade name, or trademark of such known character as to be readily identifiable with the manufacturer. Surfaces opposite stamped markings shall show no signs of penetration or unevenness.

3.7 Workmanship. Metallic surfaces shall be clean and not contain any rough or sharp edges, burrs, slivers, scratches, rough die marks, gouges, or scale. All surfaces to be welded shall be clean. Welding shall be accomplished in a manner that will prevent pits or flashes. All scale or flux deposits shall be removed from finished welds. The molded surfaces shall be free from blisters, cracks, dirt, and flash, and all gates shall be trimmed flush. The finished ice cream scoop shall not be fractured, buckled, punctured, chipped, scratched, or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be made in accordance with provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.2.2 End product inspection. The inspection lot shall consist of all the scoops of the same size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated scoop.

4.2.2.1 Visual examination. Examination shall be made of the end item for the defects listed in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish (metallic and molded surfaces, as applicable)	Evidence of discoloration, or stains.	X	
	Not smooth on all surfaces.	X	
Construction and workmanship (applicable to all components and assemblies)	Any rough or sharp edge, burr, sliver, scratch, rough die mark, gouge, pit, rust, or scale.	X	
	Any component missing.	X	
	Any component not securely attached.	X	
	Any component fractured, buckled, punctured, distorted or malformed.	X	
Bowl	Not spot welded, silver soldered, or seam welded to shank.	X	
	When spot welded, not sealed to shank.	X	
Bowl knife assembly	Knife not locked into shaft assembly.		X
	Knife broken.	X	
	Gear loose.		X
	Mechanism binds.	X	
	Torsion spring does not return knife to its original position.	X	
Handle	Not black in color.		X
	Chipped or scratched.		X
	Grooves on handle missing.	X	
	Color plug in handle base not as specified.	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding	Missing, broken, or burned through.		X
Soldering	Pinholes.		X
Identification marking	Missing, incomplete, or illegible.		X
	Surface opposite marking shows penetration.	X	
	Surface opposite marking shows unevenness.		X

4.2.2.2 Dimensional examination. Inspection shall be made of the scoop for compliance with dimensions specified herein. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. The end item shall be tested as specified in 4.3.1, 4.3.2 and 4.3.3 and failure of a test shall be recorded as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine whether the packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with defects set forth in table III. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE III. Examination of preparation for delivery requirements

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.

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TABLE III. Examination of preparation for delivery requirements (cont'd)

Examine	Defect
Workmanship	Intermediate packages not closed with tape as specified. Bulged or distorted containers.
Content per container	Number of scoops per container is more or less than required.

4.2.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and reinforcing shall be in accordance with the appendix of PPP-B-636.

4.2.4 Examination of palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored as specified in table IV. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE IV. Examination of palletization

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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4.2.5 Code and standards compliance. Proof of compliance with the requirements of NSF, as specified in 3.2, shall be submitted to the Government representative.

4.3 Tests.

4.3.1 Capacity test. The bowl shall be filled to capacity with water and the volume of the water measured in a graduated beaker to determine compliance with 3.4.1. Inability of the bowl to hold the specified volume of water shall constitute failure of the test.

4.3.2 Operational test. The thumb pad of the ice cream scoop shall be depressed to its limit and released 25 times. During and after the test, the ice cream scoop shall be inspected for conformance with the requirements specified in 3.4.2. Inability of the knife to rotate 180 degrees and failure to return without binding or malfunctioning shall constitute failure of the test.

4.3.3 Handle test. Place the open face of the ice cream scoop bowl on a flat smooth surface, the complete handle extending outward over edge of surface, clamp the bowl securely to the flat surface. At a distant on the handle 6-inches from the center line of the bowl apply a steady upward load of 30 pounds (minimum) perpendicular to the flat surface. Failure shall not occur at the joint(s), no permanent deformation shall be evident in the handle, shaft, or bowl when the load is removed. Any nonconformance with the requirements specified in 3.4.1 and 3.4.3 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each scoop shall be completely wrapped in anti-tarnish tissue paper conforming to type II, class 1 or 2 of UU-P-553. Alternatively, each scoop shall be packaged in a plastic bag fabricated from minimum 2 mil thick polyvinyl chloride film. The wrapped or bagged scoop shall then be packaged in a snug-fitting folding or metal stayed paperboard box conforming to variety 1, style II, type D or E, class e of PPP-B-566, or class 1, style C of PPP-B-665, respectively. Each box shall be securely closed as indicated by the style of the box.

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5.1.1.2 Intermediate package. Twelve scoops of one size only, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting folding or metal-stayed paperboard box conforming to variety 1, style III or VIII, type G, class 1 of PPP-B-566, or class 1, style A of PPP-B-665, respectively. Each box shall be securely closed with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, applied at the center of the length opening and along the bottom and up each side at least one-half the depth of the box. Alternately, the box shall conform to style RSC, type CF, variety SW, class domestic, grade 125 of PPP-B-636. Box closure shall be secured with the same paper tape as specified for the other boxes.

5.1.2 Level B. Packaging shall be as specified in 5.1.1.

5.1.3 Level C (commercial packaging). Scoops shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Seventy-two scoops of one size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B. Seventy-two scoops of one size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or type SF, class domestic, grade 200 of PPP-B-636. Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C (commercial packing). Scoops, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification as applicable.

5.3 Palletization (Military requirement). When specified (see 6.2), scoops of one size only, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with

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primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

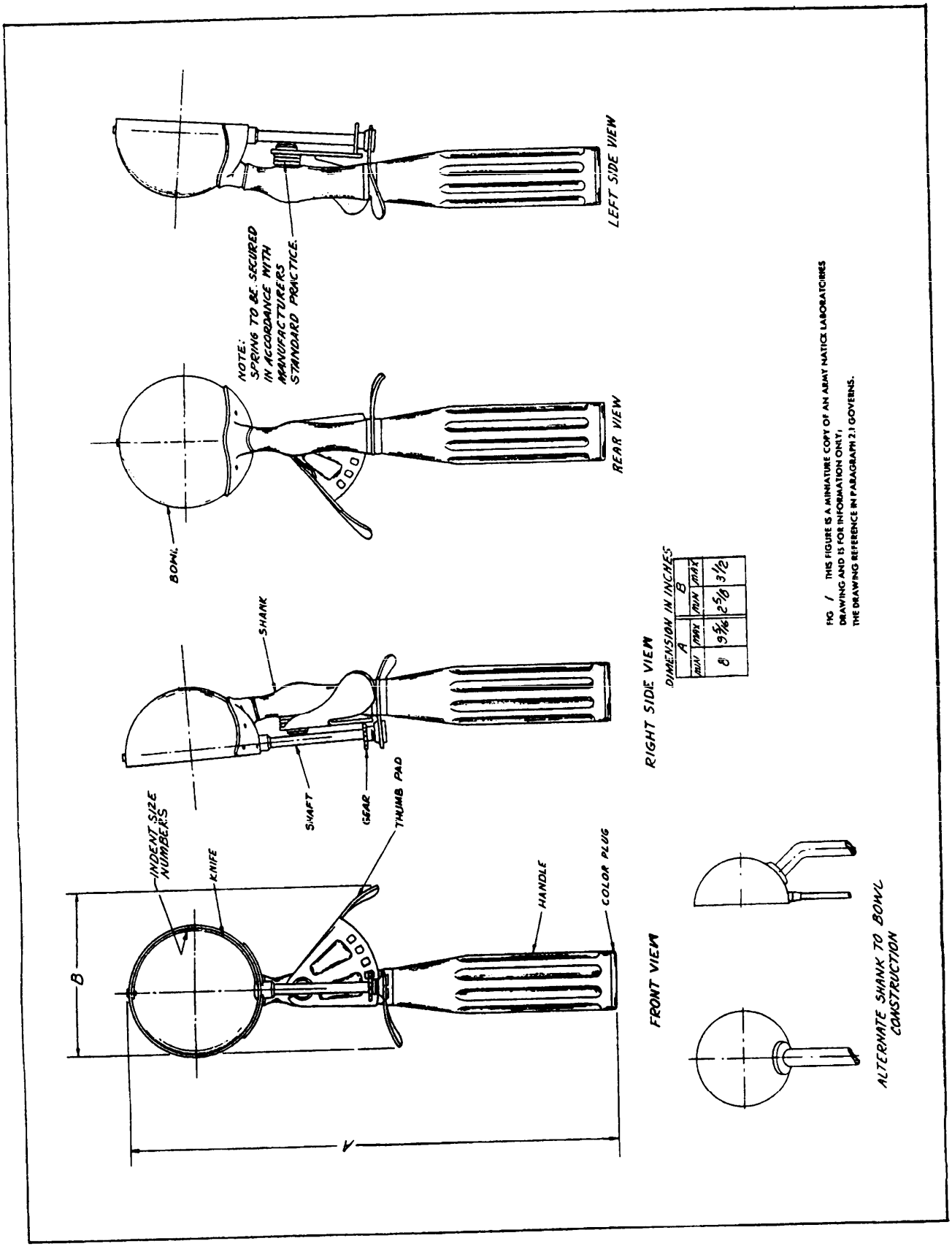
6.1 Intended use. The ice cream scoops covered by this specification are intended for dispensing and serving bulk ice cream or other soft foods as shredded, diced, or mashed vegetables, rice, et cetra.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Size of scoop required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is required (see 5.3).

6.3 Supersession data. This specification supersedes MIL-S-10897D, Scoops, Ice Cream, Mechanical, dated October 30, 1969.

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DIMENSION IN INCHES

	A		B	
	MIN	MAX	MIN	MAX
8	9 5/16		2 5/8	3 1/2

FIG 1 THIS FIGURE IS A MINIATURE COPY OF AN ARMY NATICK LABORATORIES DRAWING AND IS FOR INFORMATION ONLY; THE DRAWING REFERENCE IN PARAGRAPH 2.1 GOVERNS.

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Review activities:

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Air Force - 45

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